// ZPWR-MIDI-FX //

zpwr-midi-fx

v0.2.0

MODULE REFERENCE

wire your own MIDI processor from note-stream blocks

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MenkeTechnologies

Contents

Signal Sources

External sources you can cable into any block input or modulation route.

1 MIDI In 3 Random Mod Wheel Pitch Bend Aftertouch Velocity Expression Sustain MPE Bend MPE Pressure MPE Slide

Modulation

7 modules

Generative 3 inputs

Probabilistic melodic random walk synced to a rate.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
Range	1 – 12	3	st
Low	0 – 127	48	—
High	0 – 127	72	—
Probability	0 – 1	0.7	—

MidiLFO 3 inputs

Low-frequency modulation source (sine/tri/saw/square) on the scalar control output.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Shape	Sine · Triangle · Saw · Square		—
Depth	0 – 1	1	—

MidiEnv 3 inputs

ADSR envelope triggered by the note gate, on the scalar control output.

Param	Range	Default	Unit
Attack	1 – 5000	5	ms
Decay	1 – 5000	200	ms
Sustain	0 – 1	0.7	—
Release	1 – 5000	300	ms

MidiRandom 3 inputs

Generates random notes within a pitch range at a synced rate.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
Low Note	0 – 127	48	—
High Note	0 – 127	72	—
Gate	0 – 2	0.5	—
Velocity	1 – 127	100	—

MidiSampleHold 3 inputs

Samples and holds the incoming control value on each note.

`MidiSlew` 3 inputs

Smooths (glides) the control value over a set time.

Param	Range	Default	Unit
`Time`	0 – 2000	100	ms

`RandOctave` 3 inputs

Randomly shifts notes by up to +/- Range octaves with a chance.

Param	Range	Default	Unit
`Range`	1 – 3	1	—
`Chance`	0 – 1	1	—

Dynamics

8 modules

Velocity 3 inputs

Scales and offsets note velocity, with a modulation amount.

Param	Range	Default	Unit
Scale	0 – 2	1	—
Offset	-64 – 64	0	—
Mod Amt	-64 – 64	0	—

VelCurve 3 inputs

Reshapes velocity through a power curve.

Param	Range	Default	Unit
Curve	0.2 – 5	1	—

Humanize 3 inputs

Adds random timing and velocity jitter for a human feel.

Param	Range	Default	Unit
Timing	0 – 50	8	ms
Velocity	0 – 1	0.3	—

Accent 3 inputs

Boosts velocity on every Nth note.

Param	Range	Default	Unit
Every	1 – 16	4	—
Amount	0 – 1	0.5	—

VelClip 3 inputs

Clamps velocity between Min and Max.

Param	Range	Default	Unit
Min	1 – 127	1	—
Max	1 – 127	127	—

Ramp 3 inputs

Velocity ramp over a run of notes, ascending or descending.

Param	Range	Default	Unit
Length	2 – 16	8	—
Depth	0 – 1	0.6	—
Direction	Up · Down		—

VelCompress 3 inputs

Velocity compressor: velocities above Threshold are pulled toward it by Ratio (1 = off, 8 = hard), narrowing the dynamic range smoothly instead of hard-clamping like VelClip.

Param	Range	Default	Unit
Threshold	1 – 127	64	—
Ratio	1 – 8	2	—

VelInvert 3 inputs

Flips the velocity scale around its midpoint (soft <-> loud) by Amount, so your accents become ghost notes and the ghosts become accents - a velocity mirror, blendable to taste.

Param	Range	Default	Unit
Amount	0 – 1	1	—

Control

5 modules

CCMap 3 inputs

Remaps a CC number and scales/offsets its value.

Param	Range	Default	Unit
From	-1 – 127	1	—
To	0 – 127	11	—
Scale	0 – 2	1	—
Offset	-1 – 1	0	—

Note2CC 3 inputs

Emits a CC from each note (pitch or velocity).

Param	Range	Default	Unit
CC	0 – 127	1	—
Source	Pitch · Velocity		—

CC2Note 3 inputs

A watched CC crossing a threshold triggers a note.

Param	Range	Default	Unit
Watch	0 – 127	1	—
Note	0 – 127	60	—
Threshold	0 – 1	0.5	—

BendProc 3 inputs

Scales and/or inverts incoming pitch-bend.

Param	Range	Default	Unit
Scale	0 – 2	1	—
Invert	0 – 1	0	—

ATProc 3 inputs

Scales aftertouch, optionally converting it to a CC.

Param	Range	Default	Unit
Scale	0 – 2	1	—
To CC	-1 – 127	-1	—

Param	Range	Default	Unit

Harmony

18 modules

Chord 3 inputs

Turns each incoming note into a chord (165 types) with inversion, spread, octave doubling and strum.

Param	Range	Default
`Type`	Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major (2nd inv) · Mi	
`Inversion`	0 – 4	0
`Spread`	0 – 2	0
`Octave x2`	0 – 2	0
`Transpose`	-24 – 24	0
`Strum`	0 – 200	0

Scale 3 inputs

Quantizes incoming notes to the nearest degree of the chosen root and scale.

Param	Range	Default
`Root`	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
`Scale`	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatonic M	

Transpose 3 inputs

Shifts every note by a fixed semitone amount plus a modulatable offset.

Param	Range	Default	Unit
`Semitones`	-24 – 24	0	st
`Mod Amt`	-24 – 24	0	st

Harmonize 3 inputs

Adds up to three fixed-interval harmony voices to each note.

Param	Range	Default	Unit
`Interval 1`	-24 – 24	7	st
`Interval 2`	-24 – 24	0	st
`Interval 3`	-24 – 24	0	st

Drone 3 inputs

Doubles every note with a fixed pedal root.

Param	Range	Default	Unit
`Root`	0 – 127	36	—

`MidiCluster` 3 inputs

Adds symmetric tone-cluster voices above and below each note.

Param	Range	Default	Unit
`Width`	0 – 4	1	—
`Spread`	1 – 4	1	st

`Glissando` 3 inputs

Fills a stepwise run between consecutive notes.

Param	Range	Default	Unit
`Rate`	5 – 200	40	ms

`ChordMem` 3 inputs

Each key plays a stored chord shape (semitone intervals from node config).

`ChordProg` 3 inputs

Each note triggers the next chord in a stored root progression (node config).

`ChordVoicing` 3 inputs

Re-voices each note into a jazz shape (open / shell / quartal / spread).

Param	Range	Default	Unit
`Voicing`	Open · Shell · Quartal · Spread		—

`Octave` 3 inputs

Doubles each note up and/or down by whole octaves.

Param	Range	Default	Unit
`Up`	0 – 3	1	—
`Down`	0 – 3	0	—

`MidiFold` 3 inputs

Folds out-of-range notes back inside a Low..High window.

Param	Range	Default	Unit
`Low Note`	0 – 127	48	—
`High Note`	0 – 127	72	—

Invert 3 inputs

Mirrors pitch around a pivot note.

Param	Range	Default	Unit
Pivot	0 – 127	60	—

Diatonic 3 inputs

Transposes by scale DEGREES within Root/Scale (a third, a fifth...) instead of by raw semitones like Transpose - the shift always lands back in key. Mod Amt adds a modulatable degree offset.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatonic	
Degrees	-7 – 7	2
Mod Amt	-7 – 7	0

DiatonicChord 3 inputs

Auto-harmoniser: builds the in-key chord on every note by stacking diatonic thirds (triad, 7th...) along Root/Scale, so single-finger melodies become correct chords in the key.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatonic M	
Notes	2 – 4	3

NeoRiemann 3 inputs

Neo-Riemannian triad transforms: builds a major/minor triad on each note and applies P (parallel), L (leading-tone) or R (relative) - the maximally-smooth chord moves of late-Romantic and film harmony, each sharing two common tones with the input.

Param	Range	Default	Unit
Quality	Major · Minor		—
Transform	None · P · L · R		—

Counterpoint 3 inputs

Adds a second voice in contrary motion - when the melody rises the counter voice falls and vice-versa - quantised to Root/Scale. Interval sets the starting separation.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatoni	
Interval	1 – 24	7

Bassline 3 inputs

Follows the chord with a mono bass voice: the lowest held note dropped Octaves down, retriggered whenever the lowest note changes. Passes the original notes through and adds the bass beneath them.

Param	Range	Default	Unit
`Octaves`	1 – 3	1	—
`Velocity`	1 – 127	100	—

Probability

1 modules

Chance 3 inputs

Probabilistic gate: lets each note through with the set probability.

Param	Range	Default	Unit
`Probability`	0 – 1	0.8	—
`Mod Amt`	-1 – 1	0	—

Routing

9 modules

Split 3 inputs

Keyboard split: notes below the split point transpose by Low, the rest by High.

Param	Range	Default	Unit
Split	0 – 127	60	—
Low	-24 – 24	0	st
High	-24 – 24	0	st

Transformer 3 inputs

Rule: notes within a range are transposed and velocity-scaled.

Param	Range	Default	Unit
Low	0 – 127	0	—
High	0 – 127	127	—
Transpose	-24 – 24	0	st
VelScale	0 – 2	1	—

Merge 3 inputs

Passes both note-stream inputs through, merged into one stream.

NoteFilter 3 inputs

Passes only notes inside a key and velocity window.

Param	Range	Default	Unit
Low Note	0 – 127	0	—
High Note	0 – 127	127	—
Low Vel	1 – 127	1	—
High Vel	1 – 127	127	—

MidiLatch 3 inputs

Holds notes after release until the next note arrives.

Channel 3 inputs

Reassigns notes to a fixed MIDI channel.

Param	Range	Default	Unit
`Channel`	1 – 16	1	—

`FixedNote` 3 inputs

Forces every note to a single fixed pitch (drum / trigger).

Param	Range	Default	Unit
`Note`	0 – 127	36	—

`KeySwitch` 3 inputs

Splits the keyboard at a note; one zone passes, the other is filtered.

Param	Range	Default	Unit
`Split`	0 – 127	60	—
`Low Zone`	High passes · Low passes		—

`NoteLimit` 3 inputs

Polyphony cap: never lets more than Max notes sound at once, stealing the oldest (FIFO note-off) when a new note would exceed the limit - tames dense chords and runaway generators.

Param	Range	Default	Unit
`Max`	1 – 16	4	—

Sequencing

25 modules

Arp 3 inputs

Arpeggiator: cycles held notes through a pattern at a synced rate with gate, octaves, swing and step count. Latch holds the chord so it keeps arpeggiating after the keys are released (the next key starts a new chord).

Param	Range	Default	Unit
`Mode`	Up · Down · Up/Down · Down/Up · Converge · Diverge · As Played · Random · Chord		—
`Rate`	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
`Gate`	0 – 2	0.5	—
`Octaves`	1 – 4	1	—
`Swing`	0 – 0.75	0	—
`Steps`	1 – 16	16	—
`Latch`	0 – 1	0	—

SeqEuclid 3 inputs

Euclidean rhythm gate: spreads Pulses evenly across Steps (rotatable) at a synced rate.

Param	Range	Default	Unit
`Pulses`	1 – 16	5	—
`Steps`	1 – 16	8	—
`Rotate`	0 – 15	0	—
`Rate`	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
`Gate`	0 – 2	0.5	—

MidiGameOfLife 3 inputs

Conway's Game of Life MIDI sequencer: an 8x8 cell grid evolves under the B3/S23 rule; each clock step reads the next column and plays a note for every live cell (row maps to a scale degree from Root/Scale), and a full 8-step sweep advances one generation. Gliders, blinkers and still-lifes become ever-shifting melodic and harmonic patterns. Density sets the random fill when the grid (re)seeds. The MIDI-note counterpart to the GameOfLife CV block.

Param	Range	Default
`Root`	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
`Scale`	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatonic	
`Rate`	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
`Gate`	0 – 2	0.5
`Density`	0 – 1	0.35

MidiBrianBrain 3 inputs

Brian's Brain MIDI sequencer: an 8x8 three-state grid (ready / firing / dying) evolves under Brian's Brain rules - a ready cell ignites only with exactly two firing neighbours, firing cells pass to dying, dying cells reset. Each clock step plays a note for every firing cell in the next column (row maps to a scale degree from Root/Scale); a full 8-step sweep advances one generation. The refractory state keeps it restless and sparkly where Game of Life settles. Density sets the random fill. The MIDI-note counterpart to the BrianBrain CV block.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatonic	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0 – 2	0.5
Density	0 – 1	0.3

MidiLangton 3 inputs

Langton's Ant MIDI melody: a single turmite walks an 8x8 grid - turning right on a white cell, left on a black one, flipping each cell as it leaves, then stepping forward. Each clock advances the ant Steps cells and plays one note for its current row (mapped to a scale degree from Root/Scale), so the melody scribbles chaotically then locks into the famous periodic 'highway'. Emergent order from one trivial rule - a single moving agent, not a population.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatonic M	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0 – 2	0.5
Steps	1 – 16	1

Swing 3 inputs

Pushes off-beat eighth-notes later for a swing/shuffle feel.

Param	Range	Default	Unit
Amount	0 – 0.75	0.3	—

PatternGate 3 inputs

Gates note-ons through a 16-step on/off pattern (bitmask).

Param	Range	Default	Unit
Pattern	0 – 65535	21845	—
Steps	1 – 16	16	—

NoteOffDelay 3 inputs

Extends each note's gate by delaying its note-off.

Param	Range	Default	Unit
Delay	0 – 2000	200	ms

Ricochet 3 inputs

Accelerating bouncing retriggers with decaying velocity.

Param	Range	Default	Unit
`Bounces`	1 – 16	6	—
`Decay`	0.3 – 0.98	0.7	—
`Start`	10 – 500	100	ms

`StepSeqMidi` 3 inputs

Melodic step sequencer - semitone offsets (node config) clock the held root.

Param	Range	Default	Unit
`Rate`	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
`Gate`	0.05 – 1	0.5	—
`Velocity`	1 – 127	100	—

`Echo` 3 inputs

MIDI delay: repeats notes at a synced time with velocity feedback decay.

Param	Range	Default	Unit
`Time`	1 – 2000	250	ms
`Feedback`	0 – 0.95	0.5	—
`Repeats`	1 – 8	3	—

`Flam` 3 inputs

Precedes every note with a quiet grace note, delaying the main hit so the grace leads it - the classic drum flam.

Param	Range	Default	Unit
`Time`	1 – 200	30	ms
`Grace Vel`	0 – 1	0.5	—

`Roll` 3 inputs

Drum roll / buzz: retriggers every held note at Rate for as long as it is held, each hit gated to a fraction of the interval.

Param	Range	Default	Unit
`Rate`	1 – 50	12	Hz
`Gate`	0.05 – 0.95	0.5	—

`Trill` 3 inputs

Keyboard trill: while a note is held, alternates rapidly between it and the note Interval semitones above, at Rate.

Param	Range	Default	Unit
`Rate`	1 – 50	10	Hz
`Interval`	1 – 12	2	st
`Gate`	0.05 – 0.95	0.5	—

`Mordent` 3 inputs

One-shot ornament: each struck note plays principal -> auxiliary -> principal at the attack, then sustains, the single-flick counterpart to the continuous Trill.

Param	Range	Default	Unit
`Time`	5 – 150	40	ms
`Interval`	1 – 12	2	st
`Direction`	Upper · Lower		—

`Turn` 3 inputs

Gruppetto turn: a four-step ornament at the attack - upper auxiliary, principal, lower auxiliary, then the principal held. Time sets each step, Interval the neighbour distance.

Param	Range	Default	Unit
`Time`	5 – 150	35	ms
`Interval`	1 – 12	2	st

`Appoggiatura` 3 inputs

Leaning grace note: each struck note first sounds a long auxiliary (Interval away) that takes Time from the principal, which then sustains - the expressive long grace, unlike the quick Flam or Mordent.

Param	Range	Default	Unit
`Time`	20 – 400	120	ms
`Interval`	1 – 12	2	st
`Direction`	Upper · Lower		—

`Strum` 3 inputs

Spreads a chord's notes over time, up or down, like a guitar strum.

Param	Range	Default	Unit
`Spread`	0 – 200	20	ms
`Direction`	Up · Down		—

`MidiQuantize` 3 inputs

Snaps note starts to a synced grid by a strength amount.

Param	Range	Default	Unit
`Rate`	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
`Strength`	0 – 1	1	—

`SeqRatchet` 3 inputs

Retriggers each note Count times within a synced step.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
Count	1 – 8	4	—
Gate	0.05 – 1	0.9	—

NoteLength 3 inputs

Forces a fixed note duration (gate length).

Param	Range	Default	Unit
Length	1 – 2000	200	ms

Cascade 3 inputs

Each struck note fans out into a timed run of Steps notes climbing (or falling) the scale from it - an arpeggiated flourish fired per key, unlike Arp which cycles the whole held chord.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentator	
Steps	2 – 8	4
Time	5 – 200	50
Direction	Up · Down	

MidiTuring 3 inputs

Turing-machine sequencer (Music Thing style): a circular shift register clocks once per step and the held notes fire when the read bit is 1. Change is the probability the recycled bit flips - 0 locks an evolving loop, 1 fully randomises, between = slowly mutating melodies.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
Length	2 – 16	8	—
Change	0 – 1	0.15	—
Gate	0.05 – 1	0.5	—

Polymeter 3 inputs

Two step lanes of different lengths (LenA, LenB) clock together over the held notes, phasing against each other so the combined pattern only repeats after lcm(LenA,LenB) steps - instant polymetric movement from one chord.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
Len A	1 – 16	3	—
Len B	1 – 16	4	—
Gate	0.05 – 1	0.5	—

MidiTranceGate 3 inputs

Chops the held notes with a 16-step on/off Pattern at a synced Rate: every 'on' step retriggers all held notes, gated to a fraction of the step. The rhythmic gate behind the trance sound - pattern-driven, where Roll is a fixed pulse.

Param	Range	Default	Unit
`Rate`	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
`Pattern`	0 – 65535	21845	—
`Steps`	1 – 16	16	—
`Gate`	0.05 – 1	0.5	—

Tuning

1 modules

Microtuner 3 inputs

Stretched-octave microtuning via per-note pitch detune.

Param	Range	Default	Unit
Stretch	-50 – 50	0	cents/oct

Voicing

4 modules

MPEConvert 3 inputs

Assigns each note its own rotating channel for per-note MPE expression.

Param	Range	Default	Unit
Voices	2 – 15	15	—

Sustain 3 inputs

Sustain-pedal simulation: while Hold is engaged, note-offs are captured so the notes keep sounding; when Hold drops, every held note releases at once. Hold is a parameter, so a CC, an LFO or automation can play the pedal.

Param	Range	Default	Unit
Hold	Off · On		—

Mono 3 inputs

Monophonic note priority (last / lowest / highest).

Param	Range	Default	Unit
Priority	Last · Lowest · Highest		—

MidiUnison 3 inputs

Doubles each note into multiple voices, optionally spread across channels.

Param	Range	Default	Unit
Voices	1 – 4	2	—
Spread	Same chan · Spread chans		—